

## T3 matched EFT Lagrangian terms: EFT\_2\_after\_S1\_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S1, S2. Remaining active T3 fields: none.

### Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1$                     | $D_\mu \supset ig_1 Y B_\mu$                                       |
| $g_2$                     | $D_\mu \supset ig_2 T^A W_\mu^A$                                   |
| $g_3$                     | $D_\mu \supset ig_3 t^A G_\mu^A$                                   |
| $Y_u$                     | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$                           |
| $Y_d$                     | $-\bar{Q} Y_d H d + \text{h.c.}$                                   |
| $Y_e$                     | $-\bar{L} Y_e H e + \text{h.c.}$                                   |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F} M_F F$                                                   |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2$                   | $m_1^2 S_1^\dagger S_1$                                            |
| $m_2^2$                   | $m_2^2 S_2^\dagger S_2$                                            |
| $\lambda_1$               | $\frac{1}{2} \lambda_1 (H^\dagger H)^2$                            |
| $\lambda_{S_1}^{(1)}$     | $\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$              |
| $\lambda_{S_2}^{(1)}$     | $\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$              |
| $\lambda_{HS_1}^{(1)}$    | $\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$             |
| $\lambda_{HS_2}^{(1)}$    | $\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$             |
| $\lambda_{HS_1}^{(A)}$    | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$     |
| $\lambda_{HS_2}^{(A)}$    | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$     |
| $\lambda_5$               | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$                      |
| $\lambda_{12}^{(1)}$      | $\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$           |
| $\lambda_{12}^{(A)}$      | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$   |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2) (S_2^\dagger S_1)$      |
| $\lambda_{S_1}^{(A)}$     | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1) (S_1^\dagger T^A S_1)$  |
| $\lambda_{S_2}^{(A)}$     | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2) (S_2^\dagger T^A S_2)$  |
| $\alpha$                  | $Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$   |
| $C_2(d)$                  | $(d^2 - 1)/4$                                                      |

|                        |                                                          |
|------------------------|----------------------------------------------------------|
| $T(d)$                 | $d(d^2 - 1)/12$                                          |
| $C_{LLS_1S_2}^{ij;AB}$ | $L_i L_j S_1^A S_2^B$                                    |
| $C_{ijab}$             | $L_i L_j \phi_a \phi_b$                                  |
| $C_5^{ij}$             | $(L_i^T C L_j) H H$                                      |
| $R_W$                  | $16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$ |

## Field combinations 1 of 2

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | $H$                                                                                                                                                                                          | $\ell$                                                                                                                                  | $H$ | $\ell$ | $e$ | $F$ | $H$                                                                                                                                                                                                                     | $\ell$ |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|--------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| T3-E  | -4       | 3         | -2        | 3         | -1        | 2     | $-\frac{3}{2}$ | $-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\epsilon M_F}$       | $-\frac{\sqrt{3\hbar}\overline{y}_{1,i_3,r_1}L_{a_1,i_2}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$                   |     |        |     |     | $-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$       |        |
|       |          |           |           |           |           |       |                | $-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$                | $+\frac{3}{8}\sqrt{3\hbar}\overline{y}_{1,i_3,r_1}L_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$                          |     |        |     |     |                                                                                                                                                                                                                         |        |
|       |          |           |           |           |           |       |                | $-\frac{2\sqrt{\frac{2}{3}}\hbar\lambda_5 L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F}$ | $+\frac{\sqrt{3\hbar}\overline{y}_{2,i_3,r_1}L_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$                   |     |        |     |     | $-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\lambda_5 L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$ |        |
| cont. |          |           |           |           |           |       |                | $-\frac{2\sqrt{\frac{2}{3}}\hbar\lambda_5 L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$          | $+\frac{\sqrt{3\hbar}\overline{y}_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e}_{i_2} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ |     |        |     |     |                                                                                                                                                                                                                         |        |
| cont. |          |           |           |           |           |       |                |                                                                                                                                                                                              | $+\frac{3}{8}\sqrt{3\hbar}\overline{y}_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e}_{i_2} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$        |     |        |     |     |                                                                                                                                                                                                                         |        |
|       |          |           |           |           |           |       |                |                                                                                                                                                                                              | $+\frac{\sqrt{3\hbar}\overline{y}_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ |     |        |     |     |                                                                                                                                                                                                                         |        |
|       |          |           |           |           |           |       |                |                                                                                                                                                                                              | $+\frac{3}{8}\sqrt{3\hbar}\overline{y}_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$        |     |        |     |     |                                                                                                                                                                                                                         |        |

Field combinations 2 of 2

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | $F$ | $H$ | $\ell$                                                                                                                                                                                                        | $e$ |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|       |          |           |           |           |           |       |                |     |     | $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$                   |     |
| T3-E  | -4       | 3         | -2        | 3         | -1        | 2     | $-\frac{3}{2}$ |     |     | $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$                   |     |
|       |          |           |           |           |           |       |                |     |     | $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ |     |
| cont. |          |           |           |           |           |       |                |     |     | $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ |     |